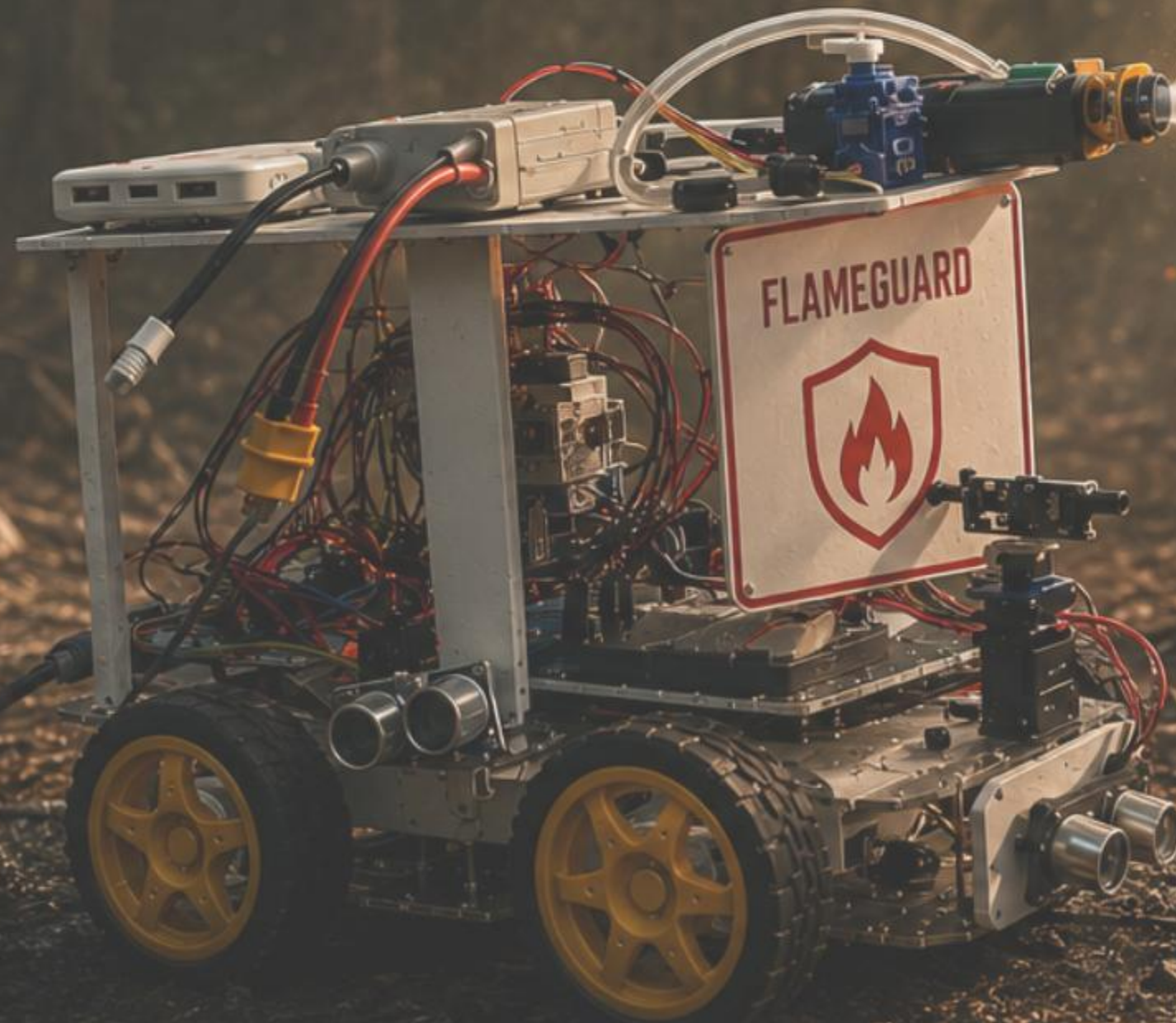


FLAMEGUARD

FIREFIGHTING ROBOT

CSE 437 — Robotics Lab | University of Asia Pacific

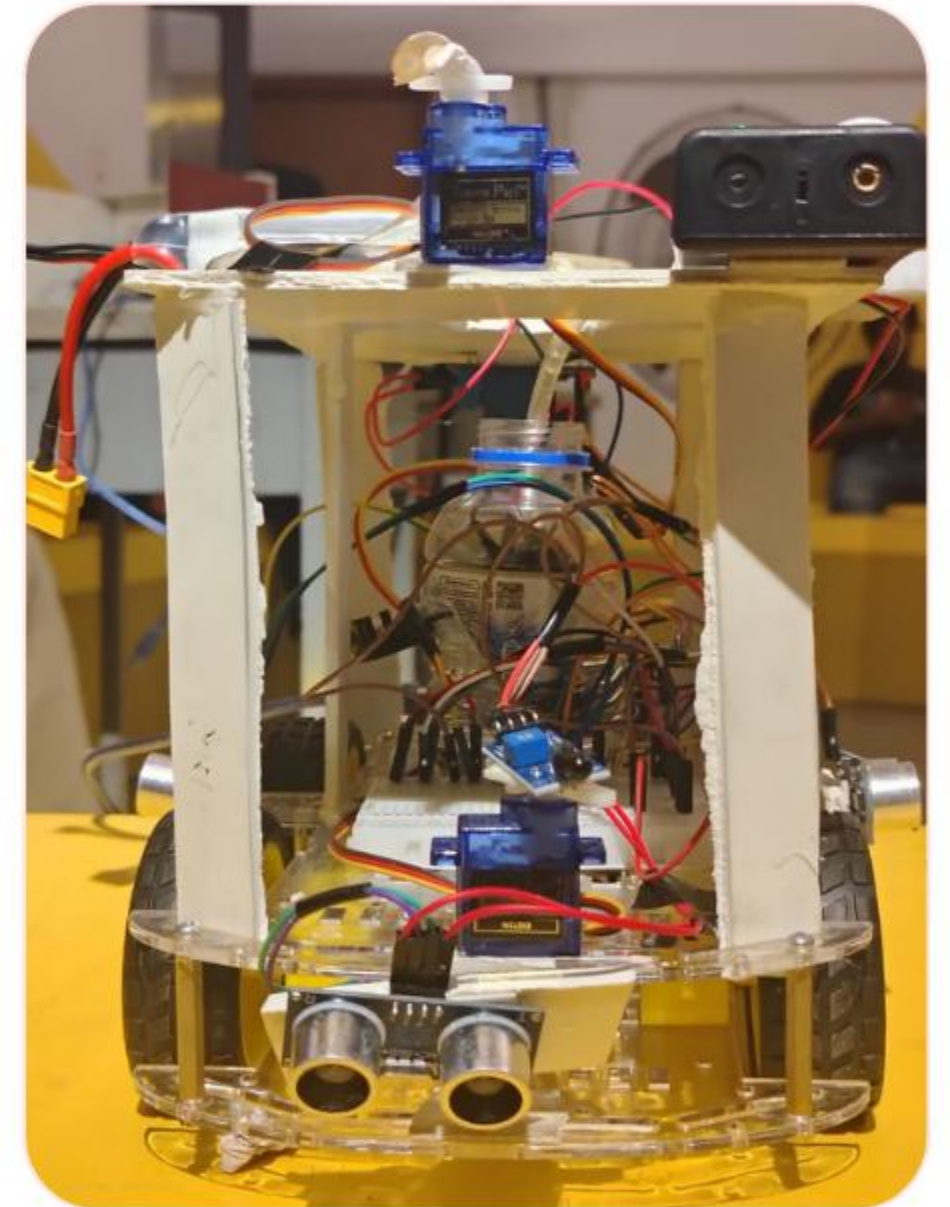


Presented By :
Lubna Akter
ID : 22201219

Presented To :
Nazmun Nahid
Assistant Professor
Department of CSE
University of Asia Pacific

Executive Summary

- Arduino based firefighting robot
- Detects and extinguishes fire automatically
- Uses flame sensor and water pump system
- Supports autonomous movement
- Reduces human risk during fire accidents
- Designed for smart emergency response

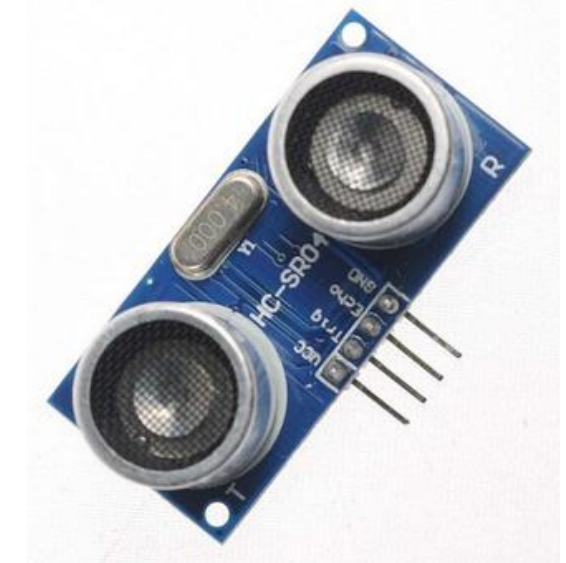
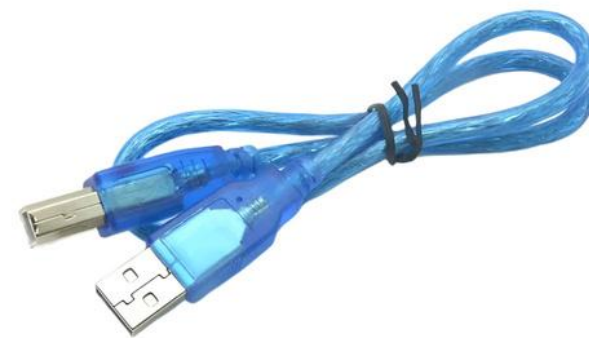
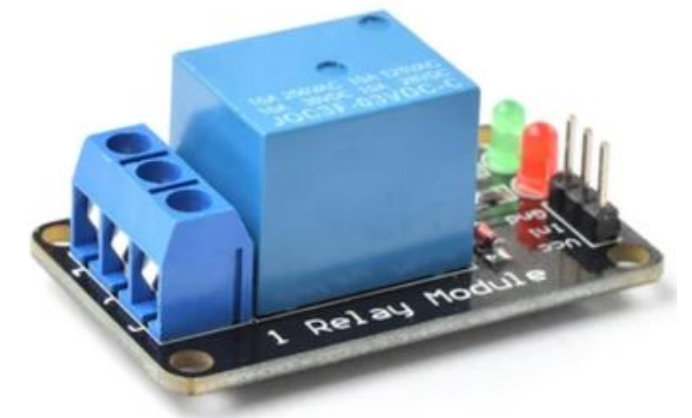
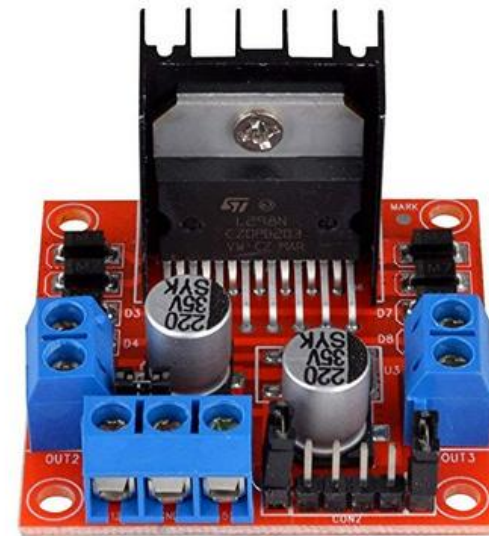


Contribution

- Conducted related study and survey
- Developed project idea and workflow
- Assisted in system planning and project design
- Collected information about firefighting robot systems.
- Collected project components and materials.



Components

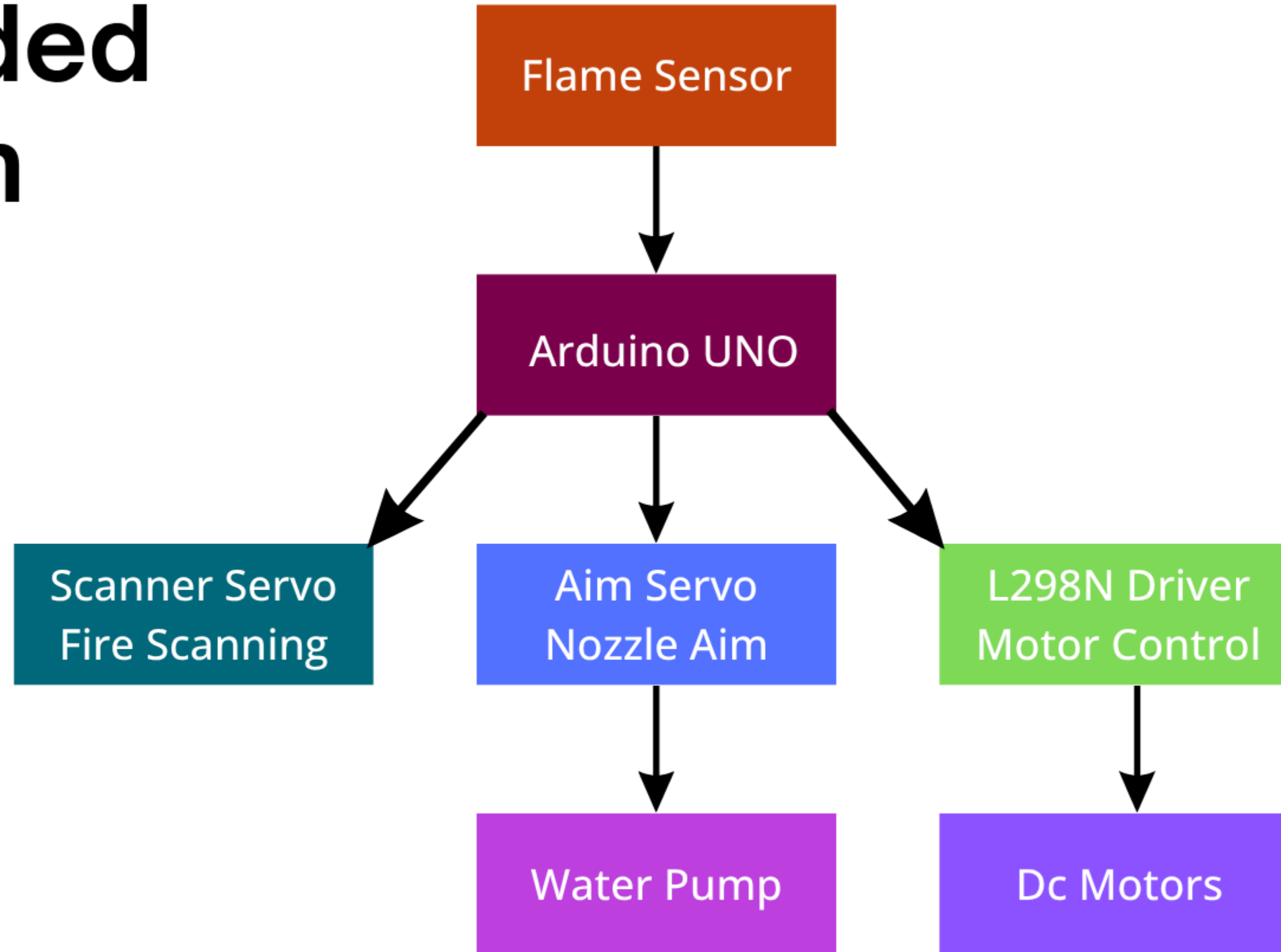


Related Study

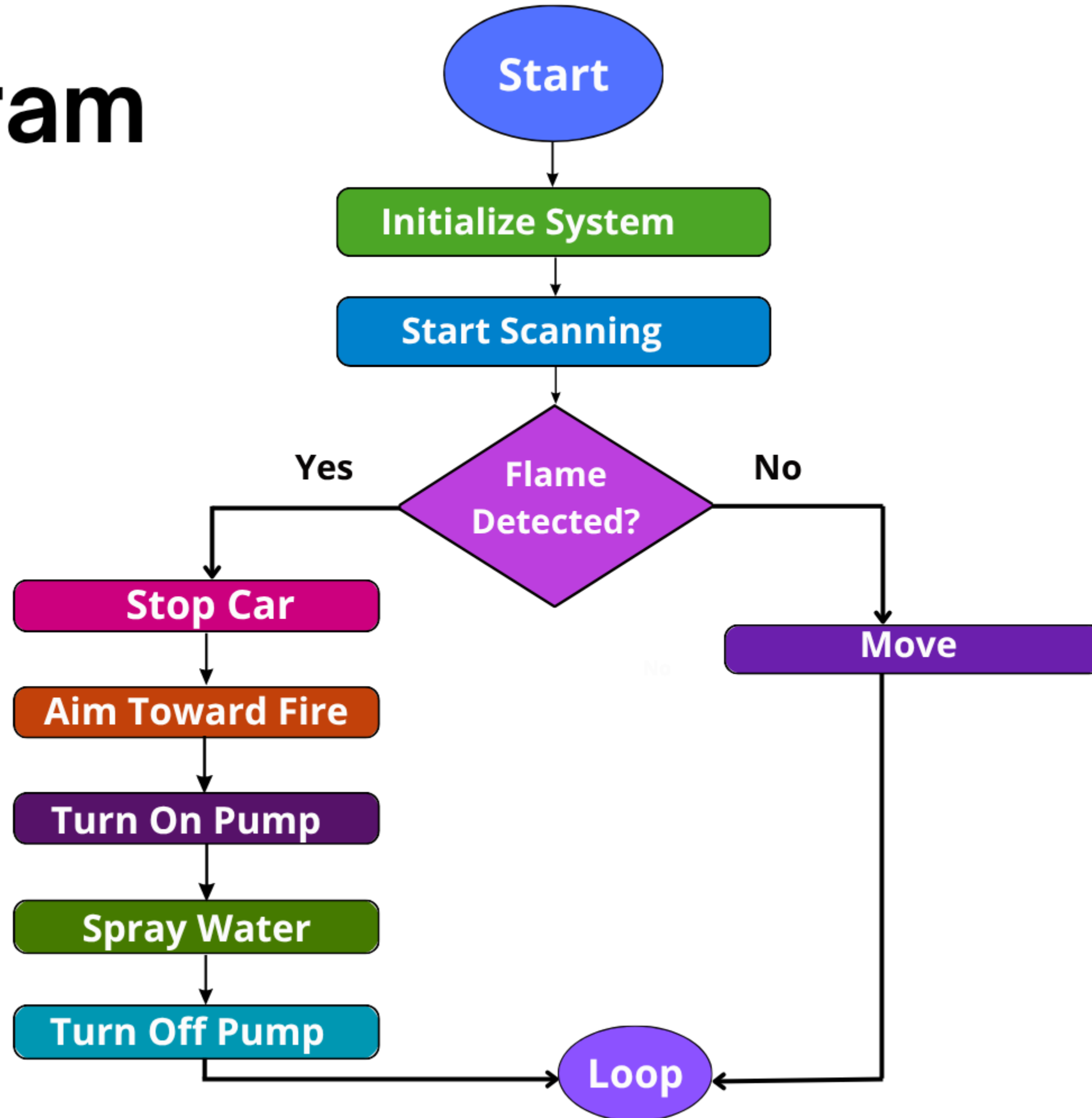
- Firefighting robots can detect and extinguish fire automatically.
(Fire Fighting Robot, 2013)
- Arduino UNO supports autonomous firefighting system.
(Wireless Firefighting Robot, 2024)
- IR and ultrasonic sensors improve obstacle avoidance accuracy.
(Develop a Multiple Interface Based Fire Fighting Robot)



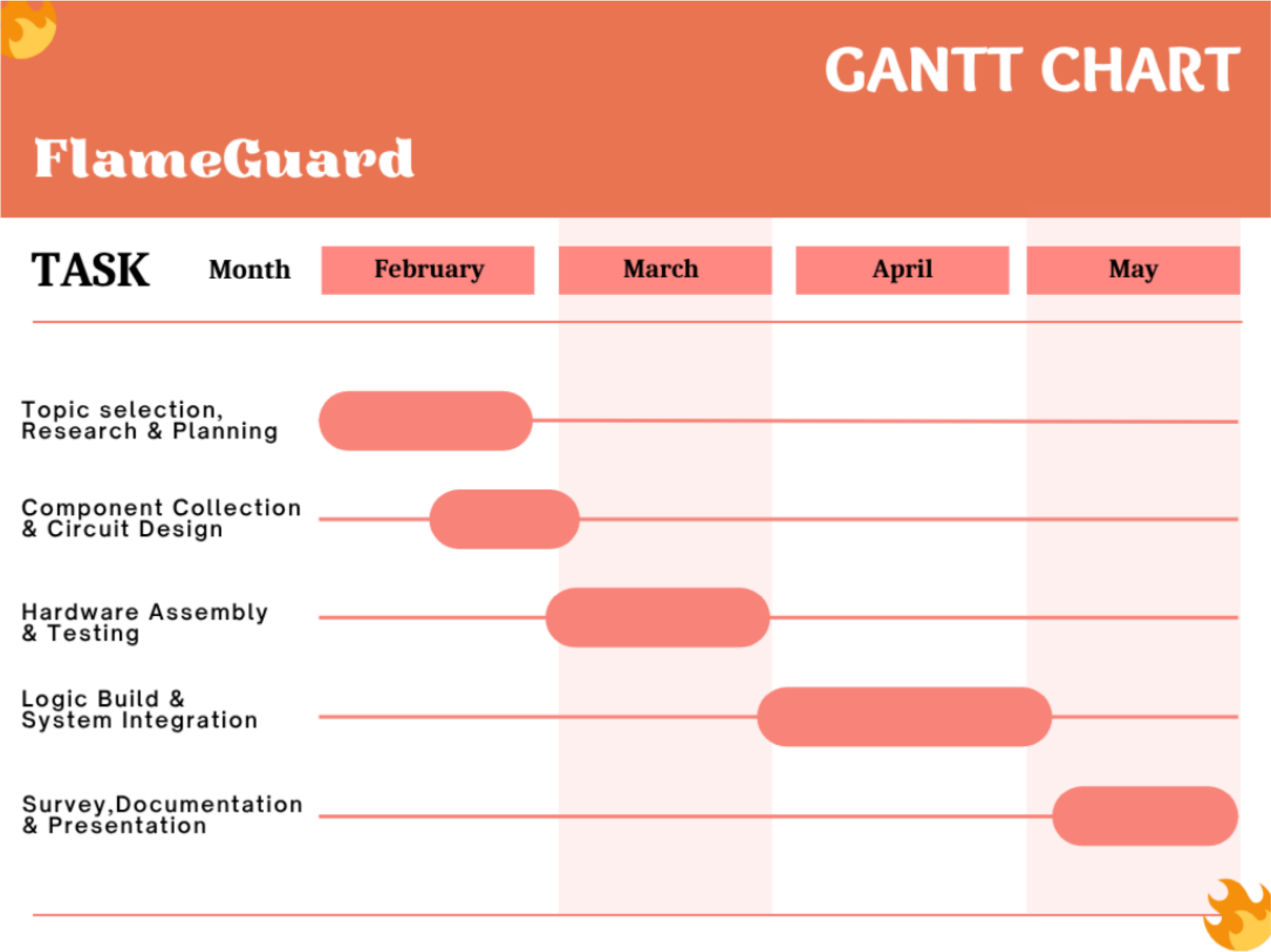
Embedded Diagram



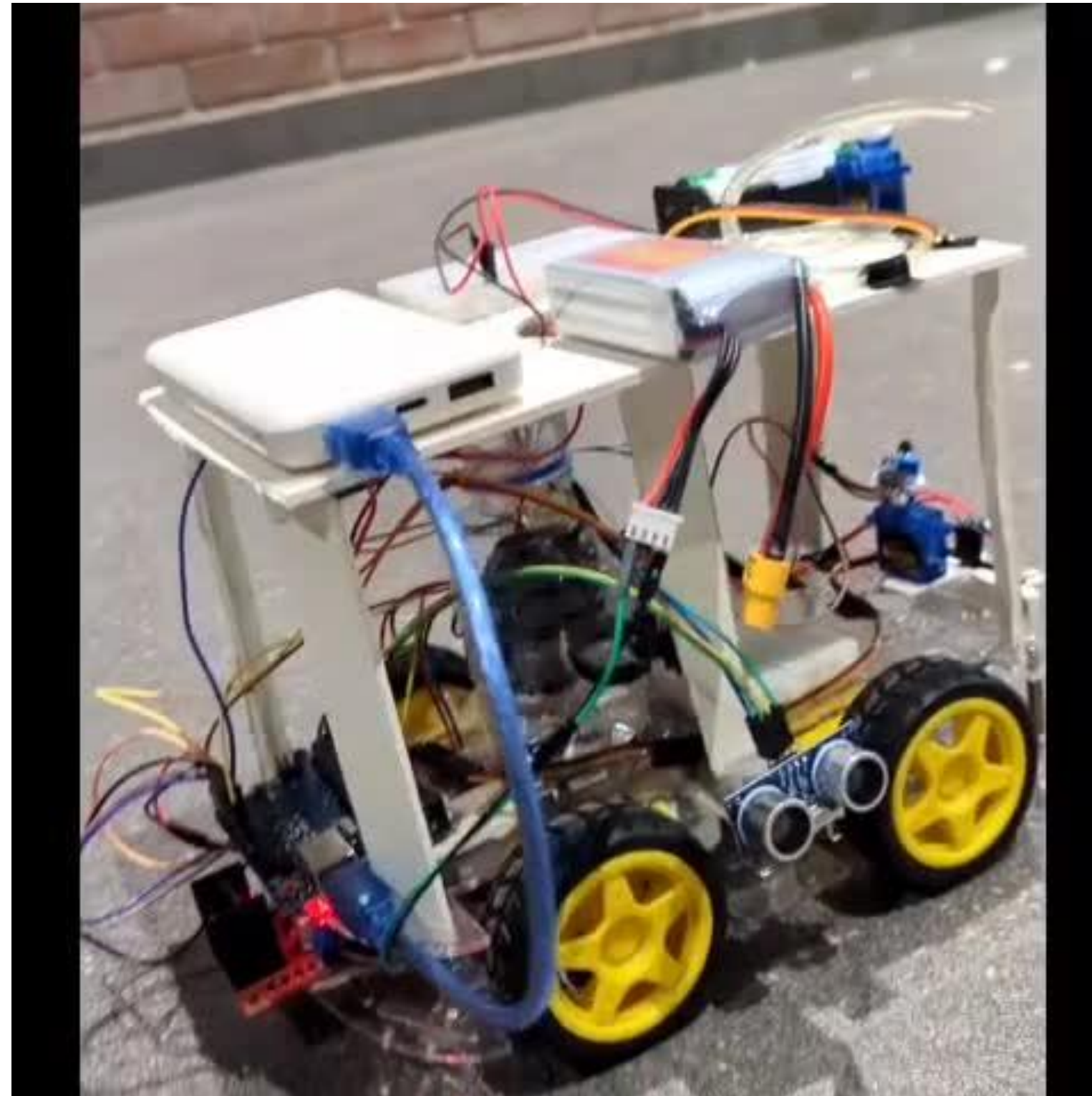
Flow Diagram

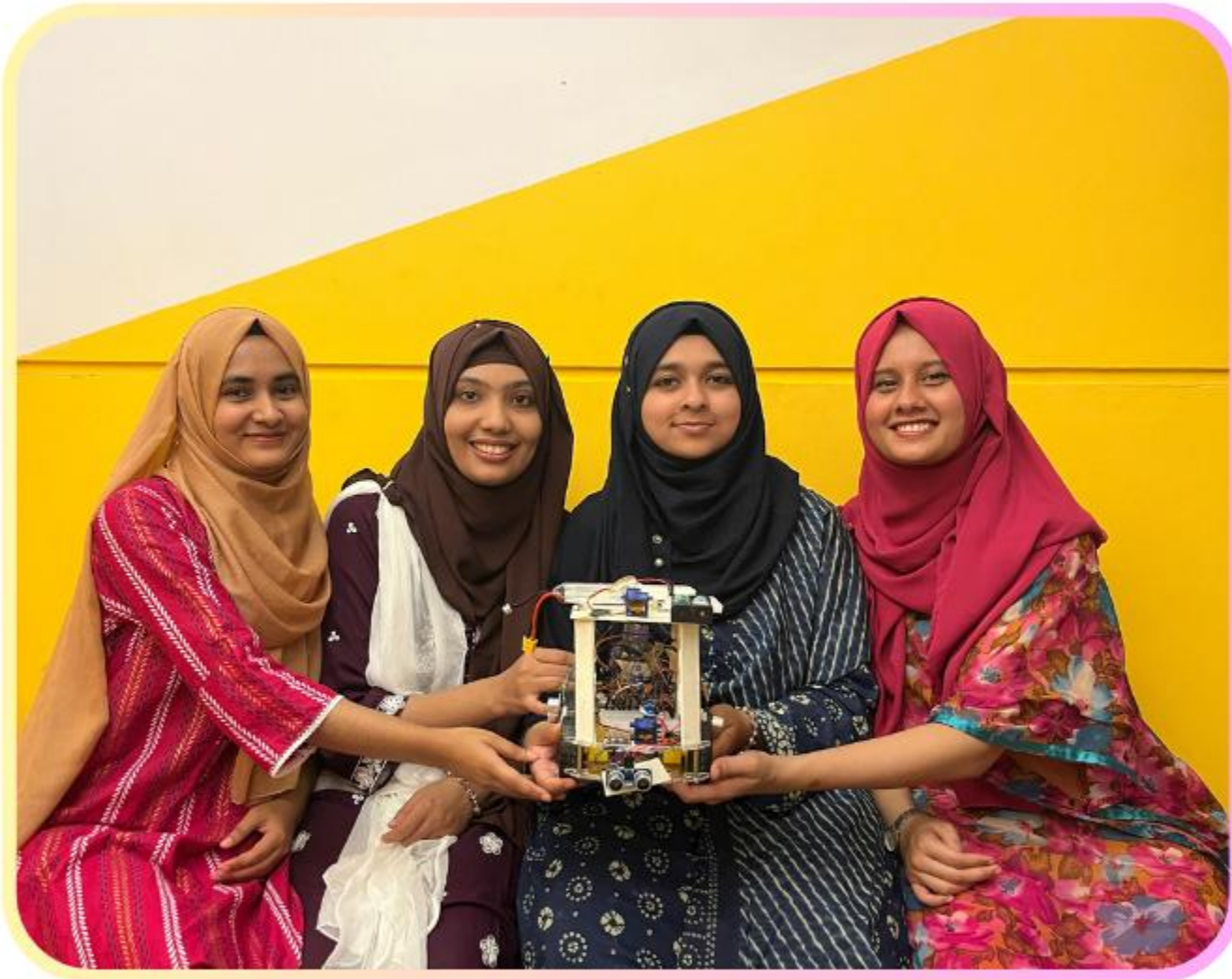


Gantt Chart



Robot Demo Video





Thank You

Any Question?